

A Facile One-pot Synthesis of α -Halo, α -allylic Aldehydes from α,α -Dihalo Ketones Utilizing Allylic Zinc Bromide

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An efficient synthesis of various α -halo, α -allylic aldehydes from α,α -dihalo ketones using both cyclic (3-bromocyclohex-1-ene zinc bromide and (Z)-3-bromocyclobut-1-ene zinc bromide) and acyclic (allylzinc bromide and cinnamylzinc bromide) type of allylic organozinc bromide with DMF as base is described. A possible reaction mechanism is also proposed.

Keywords organometallic reagent, zinc, aldehyde, allylation, rearrangement

Introduction

Multifunctional carbon is a class of compounds that contain a carbon atom bearing three or four different labile functional groups. Despite of its considerable significance in theoretical and synthetic organic chemistry, the chemistry of itself has received little attention.^[1-4] As a representative of these compounds, α -halo, α -allylic aldehydes seem especially valuable from the synthetic viewpoint. Because of their structural instability and a paucity of synthetic approaches,^[5] their potential as probes in pure and applied synthetic chemistry has yet to be exploited. Hence, the development of new and efficient synthetic methods for preparation of multifunctional carbon compounds is of particular interests and challenging.

The research in the application of allylic organometallic compounds for the synthesis of multifunctional carbon compounds is well on-going in our lab. Previously, we have reported the preparation of α -halo, α -allyl-aldehydes by reaction of allylsamarium bromide with various α,α -dihalo ketones.^[3,4] Despite of the efficiency of this method, limitation still exists since synthesis of the organosamarium compounds is relatively difficult and time-consuming.^[6] And furthermore, organosamarium compounds are considerably toxic. It comes to us that, other organometallic reagents which are easy to synthesize and less toxic while having the good efficiency should be investigated.

Zinc is a relatively non-toxic metal available at low cost. Although almost ignored for more than 100 years after their discovery, organozinc reagents today are one of the most useful class of organometallic reagents for organic and organometallic synthesis.^[7] Organozinc compounds are important intermediates in organic

chemistry because of their compatibility with many functional groups. The direct insertion of zinc into organic halides is the most attractive and simplest method for the preparation of functionalized organozinc halides.^[8]

We have recently reported the reactions of α -halo ketones with allylzinc bromide to obtain multi-product classes.^[9] Following its success, in this paper, the exploration of a new protocol for the synthesis of α -halo, α -allylic-carbonyl compounds using allylzinc bromide as well as other types of allylic zinc reagents is presented. A possible mechanism is proposed to help understanding the reactions.

Experimental

THF was distilled from sodium benzophenone under nitrogen. Et₃N, pyridine, DMF, DMSO, NMP and HMPA were distilled from CaH₂ under nitrogen. All reactions were conducted under a nitrogen atmosphere. The flash column chromatography was carried out on Merck silica gel (300–400 mesh). The IR spectra were measured on a varian 1000 FT-IR spectrometer as KBr disks (4000–400 cm⁻¹). ¹H and ¹³C NMR spectra were recorded on a Varian Mercury 400 MHz spectrometer as solutions in CDCl₃ using Me₄Si (TMS) as internal standard. Chemical shifts in ¹³C NMR spectra are reported relative to the central line of the chloroform signal (δ 77.50). High-resolution mass spectra were obtained with a GCT-TOF instrument. The starting materials were obtained by bromination, except for **11** which was prepared by published methods.^[10]

General procedure for the preparation of starting materials: α,α -dihalo ketones **1a–11**

Ketone (10 mmol) in glacial acetic acid (10 mL) was

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heated to 50 °C with stirring, and then bromine (3.5 g, 22 mmol) in glacial acetic acid (5 mL) was added dropwise. When the bromine ceased decoloration, sodium acetate (0.7 g) was added in portions. The reaction mixture was stirred for 15 min after the addition of bromine, and cooled in refrigerator for 2 h, and then poured into cold water. Finally, the crystals were filtered off, and recrystallized from 95% methyl alcohol if necessary.

General procedure for the preparation of α -halo, α -allylic-aldehydes 2a–2l & 4a–4g

Cyclic and acyclic allylic bromide (1 mmol) and zinc powder (0.0715 g, 1.1 mmol) were added into dry THF (3 mL) under a nitrogen atmosphere at room temperature. The mixture was stirred for about 15 min till zinc powder disappeared. The stirring was continued for 1 h. Then the solution of ketone substrate (0.5 mmol) was added dropwise (the reaction was monitored by TLC). The reaction mixture was stirred for 1 h and then DMF (1.6 mL) was added. The reaction mixture was stirred for further 4 h and then was poured into cold water and 5% hydrochloric acid (15 mL) was added. The resulting mixture was extracted with diethyl ether (315 mL). The organic layer was dried over anhydrous Na_2SO_4 . The solvent was removed under reduced pressure. Purification by column chromatography on silica gel afforded the products (petroleum ether as eluent).

2-Bromo-2-phenylpent-4-enal (2a) Yield 60%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.47 (s, 1H, CHO), 7.44–7.34 (m, 5H, ArH), 5.75–5.60 (m, 1H, CH), 5.11–5.05 (m, 2H, CH_2), 3.11 (d, $J=9.2$ Hz, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.37, 135.63, 132.61, 129.31, 129.29, 128.58, 120.23, 74.06, 43.39; IR (KBr) ν : 2836, 1724, 1642, 1597, 1492, 1445, 993, 924, 757, 696 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{11}^{81}\text{BrO}$ (M^+) 239.9973, found 240.0038, calcd for $\text{C}_{11}\text{H}_{11}^{79}\text{BrO}$ (M^+) 237.9993, found 238.0006.

2-Bromo-2-*p*-tolylpent-4-enal (2b) Yield 30%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.44 (s, 1H, CHO), 7.26–7.11 (m, 4H, ArH), 5.67–5.55 (m, 1H, CH), 5.05–4.99 (m, 2H, CH_2), 3.03 (d, $J=7.2$ Hz, 2H, CH_2), 2.29 (s, 3H, Me); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.41, 139.27, 132.77, 132.70, 129.92, 128.43, 120.00, 74.06, 43.22, 21.44; IR (KBr) ν : 2920, 2834, 1725, 1640, 1599, 1494, 1443, 995, 926, 812 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{12}\text{H}_{13}^{81}\text{BrO}$ (M^+) 254.0129, found 254.0130, calcd for $\text{C}_{12}\text{H}_{13}^{79}\text{BrO}$ (M^+) 252.0150, found 252.0162.

2-Bromo-2-(4-fluorophenyl)pent-4-enal (2c) Yield 52%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.46 (s, 1H, CHO), 7.44–7.06 (m, 4H, ArH), 5.72–5.63 (m, 1H, CH), 5.13–5.06 (m, 2H, CH_2), 3.10 (d, $J=3.2$ Hz, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.23, 163.14 ($^1J_{\text{CF}}=248.2$ Hz), 132.36, 131.66, 130.58 ($^3J_{\text{CF}}=8.4$ Hz), 120.53, 116.26 ($^2J_{\text{CF}}=21.7$ Hz), 73.11, 43.47; IR (KBr) ν : 2804, 1715, 1642, 1603, 1511, 1458, 998, 917, 833 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{10}\text{OF}^{81}\text{Br}$ (M^+) 257.9879, found 257.9884, calcd for $\text{C}_{11}\text{H}_{10}\text{OF}^{79}\text{Br}$ (M^+) 255.9899, found 255.9874.

2-Bromo-2-(4-chlorophenyl)pent-4-enal (2d) Yield 61%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.45 (s, 1H, CHO), 7.41–7.34 (m, 4H, ArH), 5.71–5.61 (m, 1H, CH), 5.13–5.06 (m, 2H, CH_2), 3.10–3.07 (m, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.06, 135.48, 134.33, 132.21, 130.06, 129.44, 120.66, 73.06, 43.36; IR (KBr) ν : 2833, 1732, 1642, 1592, 1492, 1432, 1013, 925, 825 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{35}\text{Cl}^{79}\text{Br}$ (M^+) 271.9604, found 271.9599, calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{35}\text{Cl}^{81}\text{Br}$ (M^+) 273.9583, found 273.9567.

2-Bromo-2-(4-bromophenyl)pent-4-enal (2e) Yield 66%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.44 (s, 1H, CHO), 7.53–7.29 (m, 4H, ArH), 5.71–5.60 (m, 1H, CH), 5.13–5.06 (m, 2H, CH_2), 3.12–3.04 (m, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 189.95, 134.84, 132.39, 132.17, 130.32, 123.68, 120.70, 73.10, 43.28; IR (KBr) ν : 2819, 1725, 1642, 1587, 1488, 1395, 1008, 925, 819 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{81}\text{Br}^{79}\text{Br}$ (M^+) 317.9078, found 317.9031, calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{81}\text{Br}_2$ (M^+) 319.9057, found 319.9020.

2-Bromo-2-*m*-tolylpent-4-enal (2f) Yield 41%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.45 (s, 1H, CHO), 7.30–7.14 (m, 4H, ArH), 5.73–5.62 (m, 1H, CH), 5.12–5.06 (m, 2H, CH_2), 3.10 (d, $J=6.8$ Hz, 2H, CH_2), 2.38 (s, 3H, Me); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.21, 138.88, 135.37, 132.52, 129.82, 128.92, 128.88, 125.41, 119.79, 73.98, 43.13, 21.67; IR (KBr) ν : 2932, 2823, 1725, 1640, 1601, 1488, 1408, 997, 922, 877, 783, 688 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{12}\text{H}_{13}\text{O}^{79}\text{Br}$ (M^+) 252.0150, found 252.0154.

2-Bromo-2-(3-chlorophenyl)pent-4-enal (2g) Yield 44%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.45 (s, 1H, CHO), 7.47–7.26 (m, 4H, ArH), 5.71–5.60 (m, 1H, CH), 5.15–5.07 (m, 2H, CH_2), 3.14–3.04 (m, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 189.62, 137.55, 135.15, 131.86, 130.21, 129.28, 128.83, 126.42, 120.46, 72.61, 43.09; IR (KBr) ν : 2834, 1727, 1639, 1571, 1478, 1417, 996, 925, 878, 785, 691 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{35}\text{Cl}^{79}\text{Br}$ (M^+) 271.9604, found 271.9612, calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{35}\text{Cl}^{81}\text{Br}$ (M^+) 273.9583, found 273.9604.

2-Bromo-2-(3-bromophenyl)pent-4-enal (2h) Yield 45%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.44 (s, 1H, CHO), 7.62–7.26 (m, 4H, ArH), 5.69–5.62 (m, 1H, CH), 5.15–5.07 (m, 2H, CH_2), 3.13–3.05 (m, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 189.56, 137.72, 132.18, 131.79, 131.57, 130.40, 126.88, 123.16, 120.43, 72.49, 43.05; IR (KBr) ν : 2836, 1725, 1640, 1564, 1474, 1416, 995, 925, 879, 783, 691 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{79}\text{Br}^{81}\text{Br}$ (M^+) 317.9078, found 317.9071.

2-Bromo-2-(naphthalen-2-yl)pent-4-enal (2i) Yield 54%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.57 (s, 1H, CHO), 7.87–7.50 (m, 7H, ArH), 5.75–5.64 (m, 1H, CH), 5.12–5.07 (m, 2H, CH_2), 3.21 (d, $J=6.4$ Hz, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.55, 133.46, 133.25, 133.10, 132.66, 129.31, 128.81, 128.08, 127.94, 127.67, 127.37, 125.78, 120.35, 74.24,

43.39; IR (KBr) ν : 2833, 1725, 1636, 1598, 1505, 1401, 1130, 923, 815, 746 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{15}\text{H}_{13}\text{O}^{79}\text{Br}$ (M^+) 288.0150, found 288.0147, calcd for $\text{C}_{15}\text{H}_{13}\text{O}^{81}\text{Br}$ (M^+) 290.0129, found 290.0148.

2-Bromo-2-(naphthalen-1-yl)pent-4-enal (2j)

Yield 52%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.63 (s, 1H, CHO), 8.00–7.24 (m, 7H, ArH), 5.72–5.61 (m, 1H, CH), 5.05–4.97 (m, 2H, CH_2), 3.48–3.43 (m, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 191.18, 134.65, 132.38, 131.35, 130.86, 130.05, 129.74, 128.94, 126.93, 126.21, 125.18, 124.67, 119.84, 75.83, 42.18; IR (KBr) ν : 2836, 1722, 1635, 1597, 1510, 1397, 991, 924, 775 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{15}\text{H}_{13}\text{O}^{79}\text{Br}$ (M^+) 288.0150, found 288.0148.

2-Chloro-2-(4-chlorophenyl)pent-4-enal (2k)

Yield 40%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.39 (s, 1H, CHO), 7.40–7.35 (m, 4H, ArH), 5.70–5.61 (m, 1H, CH), 5.13–5.08 (m, 2H, CH_2), 3.01 (d, $J=6.8$ Hz, 2H, CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 191.92, 135.47, 134.09, 131.26, 129.48, 129.08, 120.83, 77.18, 42.91; IR (KBr) ν : 2723, 1732, 1640, 1595, 1492, 1400, 1012, 924, 824 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{10}^{35}\text{Cl}_2\text{O}$ (M^+) 228.0109, found 228.0112, calcd for $\text{C}_{11}\text{H}_{10}^{35}\text{Cl}^{37}\text{ClO}$ (M^+) 230.0079, found 230.0141.

2-Allyl-2-bromooctanal (2l) Yield 41%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.39 (s, 1H, CHO), 5.84–5.74 (m, 1H, CH), 5.21–5.17 (m, 2H, CH_2), 2.77 (t, $J=6.8$ Hz, 2H, CH_2), 2.01–1.91 (m, 2H, CH_2), 1.29–1.45 (m, 8H, 4 CH_2), 0.88 (t, $J=7.2$ Hz, 3H, Me); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 193.71, 132.18, 119.98, 73.07, 40.46, 36.41, 31.59, 29.36, 25.28, 22.64, 14.15; IR (KBr) ν : 2956, 2929, 2859, 1728, 1463, 1378, 995, 924 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{11}\text{H}_{19}^{81}\text{BrO}$ (M^+) 248.0599, found 248.0592, calcd for $\text{C}_{11}\text{H}_{19}^{79}\text{BrO}$ (M^+) 248.0619, found 248.0674.

1,1-Dibromo-2-phenylpent-4-en-2-ol (3) Yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 7.49–7.33 (m, 5H, ArH), 6.00 (s, 1H, CH), 5.58–5.46 (m, 1H, CH), 5.18–5.08 (m, 2H, CH_2), 2.94 (d, $J=6.8$ Hz, 2H, CH_2), 2.78 (s, 1H, OH); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 140.46, 132.28, 128.50, 128.26, 126.47, 120.45, 78.87, 57.49, 43.78; IR (KBr) ν : 3440, 2915, 1595, 1497, 1440, 962, 913, 743, 714 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_8\text{H}_7^{79}\text{Br}^{81}\text{BrO}$ ($\text{M}^+ - \text{C}_3\text{H}_5$) 278.8843, found 278.8813.

2-Bromo-2-(cyclohex-2-enyl)-2-phenylacetaldehyde (4a)

Yield 32%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.46 (s, 1H, CHO), 7.50–7.33 (m, 5H, ArH), 5.77–5.73 (m, 1H, CH), 5.45–5.40 (m, 1H, CH_2), 3.22–3.29 (m, 1H, CH), 2.05–1.52 (m, 6H, 3 CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 191.09, 134.71, 130.99, 129.61, 129.16, 128.98, 126.56, 81.71, 44.05, 26.80, 25.42, 22.26; IR (KBr) ν : 2836, 1724, 1642, 1597, 1532, 1445, 987, 913, 752, 696 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{14}\text{H}_{15}^{79}\text{BrO}$ (M^+) 278.0306, found 278.0304, calcd for $\text{C}_{14}\text{H}_{15}^{81}\text{BrO}$ (M^+) 280.0286, found 280.0255.

2-Bromo-2-(4-chlorophenyl)-2-(cyclohex-2-enyl)-acetaldehyde (4b) Yield 30%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.45 (s, 1H, CHO), 7.45–7.34 (m,

4H, ArH), 5.81–5.71 (m, 1H, CH), 5.42–5.38 (m, 1H, CH_2), 3.26–3.18 (m, 1H, CH), 2.10–1.41 (m, 6H, 3 CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.68, 135.08, 133.14, 131.30, 130.84, 128.85, 128.85, 125.89, 80.36, 43.89, 26.42, 25.17, 21.93; IR (KBr) ν : 2936, 1726, 1639, 1591, 1539, 1491, 1014, 902, 824 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{14}\text{H}_{14}\text{O}^{79}\text{Br}^{35}\text{Cl}$ (M^+) 311.9917, found 311.9914.

2-Bromo-2-(4-bromophenyl)-2-(cyclohex-2-enyl)-acetaldehyde (4c)

Yield 35%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.45 (s, 1H, CHO), 7.52–7.35 (m, 4H, ArH), 5.82–5.71 (m, 1H, CH), 5.45–5.38 (m, 1H, CH_2), 3.25–3.15 (m, 1H, CH), 2.03–1.44 (m, 6H, 3 CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.57, 133.69, 131.78, 131.28, 131.09, 125.84, 123.29, 80.36, 43.84, 26.39, 25.13, 21.90; IR (KBr) ν : 2939, 1725, 1636, 1584, 1488, 1396, 1009, 916, 821 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{14}\text{H}_{14}\text{O}^{81}\text{Br}^{79}\text{Br}$ (M^+) 357.9391, found 357.9389, calcd for $\text{C}_{11}\text{H}_{10}\text{O}^{81}\text{Br}_2$ (M^+) 359.9370, found 359.9350.

2-Bromo-2-(cyclooct-2-enyl)-2-phenylacetaldehyde (4d)

Yield 27%; colorless oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.42 (s, 1H, CHO), 7.48–7.29 (m, 5H, ArH), 5.63–5.60 (m, 1H, CH), 5.35–5.30 (m, 1H, CH_2), 3.50–3.60 (m, 1H, CH), 2.27–1.31 (m, 10H, 5 CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 191.08, 135.08, 130.88, 129.14, 129.10, 128.81, 128.75, 82.12, 43.21, 33.73, 29.40, 26.89, 26.82, 25.60; IR (KBr) ν : 2846, 1728, 1640, 1595, 1511, 1449, 989, 914, 764, 695 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{16}\text{H}_{19}^{79}\text{BrO}$ (M^+) 306.0619, found 306.0617, calcd for $\text{C}_{16}\text{H}_{19}^{81}\text{BrO}$ (M^+) 308.0599, found 308.0580.

2-Bromo-2-(4-bromophenyl)-2-(cyclooct-2-enyl)-acetaldehyde (4e)

Yield 30%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.40 (s, 1H, CHO), 7.50–7.33 (m, 4H, ArH), 5.64–5.61 (m, 1H, CH), 5.31–5.25 (m, 1H, CH_2), 3.52–3.46 (m, 1H, CH), 2.24–1.31 (m, 10H, 5 CH_2); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 191.01, 134.32, 131.92, 131.44, 130.84, 128.77, 123.27, 81.48, 43.26, 33.82, 29.42, 26.94, 25.61; IR (KBr) ν : 2843, 1711, 1640, 1593, 1504, 1441, 986, 907, 826 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{16}\text{H}_{18}\text{O}^{79}\text{Br}_2$ (M^+) 383.9724, found 383.9722, calcd for $\text{C}_{16}\text{H}_{18}\text{O}^{81}\text{Br}^{79}\text{Br}$ (M^+) 385.9704, found 385.9665.

2-Bromo-2,3-diphenylpent-4-enal (4f) $dr=3:5$.

Yield 23%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.55–9.42 (m, 1H, CHO), 7.40–6.90 (m, 10H, ArH), 6.31–5.88 (m, 1H, CH), 5.32–5.10 (m, 2H, CH_2), 4.45–4.35 (m, 1H, CH); ^{13}C NMR (CDCl_3 , 100 MHz) δ : 190.07, 189.47, 138.43, 138.13, 137.36, 136.13, 133.84, 130.61, 130.24, 129.43, 129.00, 128.83, 128.26, 128.18, 127.92, 127.85, 127.50, 127.24, 126.17, 119.47, 118.93, 77.90, 57.31, 56.23; IR (KBr) ν : 2738, 1702, 1639, 1593, 1553, 1484, 996, 904, 779, 698 cm^{-1} ; HRMS (EI^+) calcd for $\text{C}_{16}\text{H}_{18}\text{O}^{81}\text{Br}$ (M^+) 316.0286, found 316.0255.

2-Bromo-2-(4-bromophenyl)-3-phenylpent-4-enal (4g) $dr=3:2$. Yield 25%; yellow oil. ^1H NMR (CDCl_3 , 400 MHz) δ : 9.50–9.43 (m, 1H, CHO), 7.48–

6.88 (m, 9H, ArH), 6.23–5.92 (m, 1H, CH), 5.28–5.14 (m, 2H, CH₂), 4.35 (d, $J=8.4$ Hz, 1H, CH); ¹³C NMR (CDCl₃, 100 MHz) δ : 189.77, 189.05, 137.87, 136.82, 135.67, 132.99, 132.02, 131.35, 131.27, 131.19, 130.53, 130.14, 129.45, 128.82, 128.50, 128.11, 127.76, 127.56, 123.44, 123.26, 119.97, 119.42, 78.86, 57.31, 56.30; IR (KBr) ν : 2817, 1713, 1636, 1585, 1490, 1448, 992, 901, 834, 778, 696 cm⁻¹; HRMS (EI⁺) calcd for C₁₇H₁₄O⁷⁹Br₂ (M⁺) 391.9411, found 391.9409, calcd for C₁₇H₁₄O⁸¹Br⁷⁹Br (M⁺) 393.9391, found 393.9382.

General procedure for the preparation of cyclic and acyclic allylic bromide

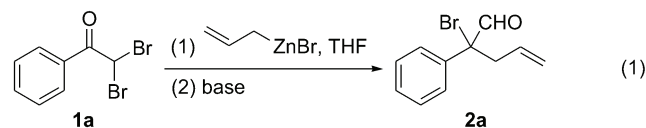
Cinnamyl bromide This compound was prepared according to a reported procedure^[11] with modification for the purification. Triphenylphosphine (5.32 g, 20.4 mmol) was dissolved in 20 mL of dry acetonitrile, and Br₂ (3.2 g, 20 mmol) was added dropwise over a 15 min period with cooling to maintain the solution at 0–10 °C. The solution was allowed to warm to room temperature, and then cinnamyl alcohol (2.7 g, 20 mmol) was added in acetonitrile (5 mL) whereupon an exotherm occurred ($T=32$ °C). The solution was reacted for 1 h and then was poured into 500 mL of ether. Triphenylphosphine oxide precipitated from this solution upon standing overnight. The solution was filtered, and the solvent was removed by evaporation under reduced pressure. Purification by column chromatography on silica gel afforded the product (300–400 mesh, petroleum ether and ethyl acetate as eluent). Yield 87%; yellow oil. ¹H NMR (CDCl₃, 400 MHz) δ : 7.41–7.26 (m, 5H, ArH), 6.70–6.62 (m, 1H, CH), 6.46–6.37 (m, 1H, CH), 4.17 (d, $J=8.0$ Hz, 2H, CH₂).

(Z)-3-bromocyclooct-1-ene This compound was prepared according to an earlier reported procedure.^[12] A mixture of the substrate (20 mL, 154 mmol), NBS (27.4 g, 154 mmol), and benzoyl peroxide (70%, 100 mg, 0.29 mmol) in CCl₄ (100 mL) was heated at reflux for 1 h. The reaction mixture was cooled to 0 °C, and then filtered. The filtrate was washed with 5% aq. NaHCO₃ then dried and concentrated. Purification via fractional distillation at reduced pressure gave product. Yield 61%; colourless oil; b.p. 68–70 °C (400 Pa). ¹H NMR (CDCl₃, 400 MHz) δ : 5.82–5.74 (m, 1H, CH), 5.65–5.55 (m, 1H, CH), 5.00–4.90 (m, 1H, CH), 2.30–1.24 (m, 10H, 5CH₂).

3-Bromocyclohex-1-ene This compound was prepared according to an earlier reported procedure.^[13] To a suspension of AIBN and *N*-bromosuccinimide (NBS, 23.1 g, 0.13 mol) in tetrachloromethane (75 mL), cyclohexene (20 mL, 0.2 mol) was added under argon. The mixture was refluxed until the disappearance of NBS. The mixture was then filtered, concentrated and distilled under vacuum. The liquid was stored in the absence of light. Yield 65%; colourless oil; b.p. 63–64 °C (2000 Pa). ¹H NMR (CDCl₃, 400 MHz) δ : 5.94–5.90 (m, 1H, CH), 5.85–5.80 (m, 1H, CH), 4.86–4.84 (m, 1H, CH), 2.30–1.64 (m, 6H, 3CH₂).

Results and Discussion

The reaction of 2,2-dibromo-1-phenylethanone (**1a**) and allylzinc bromide was selected as model reaction to optimize the reaction conditions (Eq. 1).



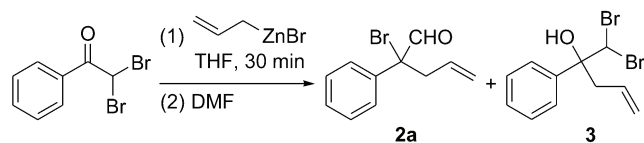
As summarized in Table 1, after reacted at room temperature for different time, 2-bromo-2-phenylpent-4-enal (**2a**) was obtained in low to moderate yield. The product was not obtained without base (Entry 1) or with the weak base K₂CO₃ (Entry 2). Some other bases such as NaOH, CaH₂, NaOEt, Py or NEt₃ were also examined later and all gave discouraging results again, with yields no more than 30% (Entries 3–7). Our further investigation showed better results on DMF, HMPA, NMP and DMSO, which gave moderate yields ranging from 35% to 60% (Entries 8–11). It is worth to note that the use of DMF as a base yields the best results among all the bases used here (60%, Entry 8). We next optimized the amount of DMF (Entries 8, 12, 13) and it was found that 20 equiv. of DMF is essential to promote this reaction. Based on these results, DMF (20 equiv.) was utilized as base hereafter.

We then optimize the optimal amount of allylic zinc bromide used in this reaction. The substrate/allylzinc bromide ratio was explored in an increased manner for optimizing the yield of **2a**. From the results in Table 2, one can easily see that the reaction carried out by using

Table 1 Comparison of the effect of bases in the reaction of α,α -dibromo-1-phenylethanone (**1a**) with allylzinc bromide^a

Entry	Base	mmol	Time/h	Yield ^b /%
1	None	10	12	0
2	K ₂ CO ₃	10	12	0
3	NaOH	10	12	12
4	CaH ₂	10	12	22
5	NaOEt	10	12	27
6	Py ^c	10	12	10
7	NEt ₃ ^c	10	12	12
8	DMF ^c	10	4	60
9	HMPA ^c	10	4	47
10	NMP ^c	10	4	51
11	DMSO ^c	10	12	35
12	DMF ^c	5	4	34
13	DMF ^c	15	4	55

^a Reaction conditions: 2,2-dibromo-1-phenylethanone (0.5 mmol), allylzinc bromide (1.0 mmol), THF (3 mL) at room temperature under nitrogen. ^b Isolated yield is based on **1a** after silica gel chromatography. ^c Distilled from sodium or CaH₂ under nitrogen.

Table 2 Influence of the substrate/allylzinc bromide ratio on the reaction of **2a**^a

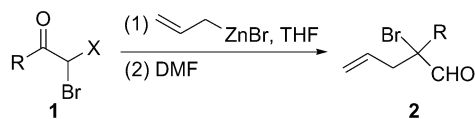
Entry	Sub./allylzinc bromide	Yield of 2a ^b /%	Yield of 3 ^b /%
1	1 : 1	25	64
2	1 : 1.5	37	54
3	1 : 2	60	25
4	1 : 2.5	58	26

^a Reaction conditions: substrate (0.5 mmol), allylzinc bromide, THF (3 mL), DMF (1.6 mL) at room temperature under nitrogen for 4 h. ^b Isolated yield is based on substrate after silica gel chromatography.

stoichiometric amounts of allylzinc bromide afforded compound **2a** in 25% yield and compound **3** in 64% yield (Entry 1). With the increase of the ratio, the highest yield of compound **2a** was obtained when 2 equiv. allylzinc bromide was used (Entry 3).

To investigate the generality of our method, a number of α,α -dihalo-ketones were further subjected to this reaction under the optimized conditions and the results are summarized in Table 3.

These results clearly show that the substrates with aromatic rings bearing electroneutral (Entry 1), electron-

Table 3 Reactions of α,α -dihalo-ketones with allylzinc bromide^a

Entry	R	X	Product	Yield ^b /%
1	C ₆ H ₅ (1a)	Br	2a	60
2	<i>p</i> -CH ₃ C ₆ H ₄ (1b)	Br	2b	30
3	<i>p</i> -FC ₆ H ₄ (1c)	Br	2c	52
4	<i>p</i> -ClC ₆ H ₄ (1d)	Br	2d	61

Continued				
Entry	R	X	Product	Yield ^b /%
5	<i>p</i> -BrC ₆ H ₄ (1e)	Br	2e	66
6	<i>m</i> -CH ₃ C ₆ H ₄ (1f)	Br	2f	41
7	<i>m</i> -ClC ₆ H ₄ (1g)	Br	2g	44
8	<i>m</i> -BrC ₆ H ₄ (1h)	Br	2h	45
9	2-Naphthyl (1i)	Br	2i	54
10	1-Naphthyl (1j)	Br	2j	52
11	<i>p</i> -ClC ₆ H ₄ (1k)	Cl	2k	40
12	CH ₃ (CH ₂) ₅ (1l)	Br	2l	41

^a Reaction conditions: α,α -dihalo-ketones (0.5 mmol), allylzinc bromide (1.0 mmol), THF (3 mL), DMF (1.6 mL) at room temperature under nitrogen for 4 h. ^b Isolated yield is based on α,α -dihalo-ketones after silica gel chromatography.

withdrawing (Entries 3–5, 7, 8) or electron-donating (Entries 2, 6) substituents could undergo reactions smoothly to afford the desired products with moderate yields (30%–66%). The results also indicate that electronic natures have a great impact on the processes. Electron-deficient groups at the *para* position of phenyl ring were well tolerated in better yields (Entries 3–5). A substituent at *meta* position, whether electron-donating or electron-withdrawing, had almost no influence on the reactions (Entries 6–8). Examination of the results reveals that the steric effect (Entries 9, 10)

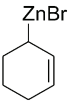
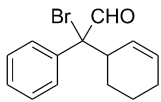
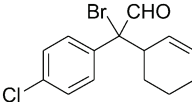
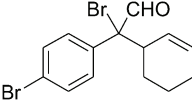
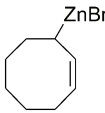
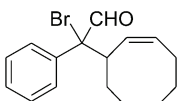
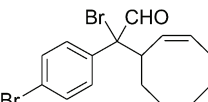
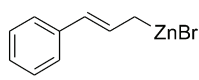
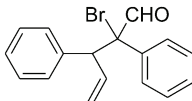
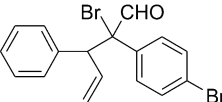
played a minimal role in governing the yields of the reactions. When X=Cl (Entry 11), the reaction could provide the desired product in 40% yield. The decreased yield compared to Entry 4 is probably due to that Br is more reactive than Cl. Besides, an aliphatic substrate could also be converted to the corresponding α -halo, α -allyl-aldehyde (Entry 12).

In order to further determine the scope of the method, other types of allylic zinc reagents^[14] were also investigated (Table 4). Unfortunately, utilizing the same experimental protocol used for **1a**, the reactions showed

lower yields. Although the yields ranging from 23% to 35% are lower than that for allylzinc bromide (60%), further optimization of reaction conditions is strongly encouraged to ensure higher yields. In general, we are able to expand the scope of our method to at least four types of allylic zinc reagents, in the context that higher yields are yet to be chased.

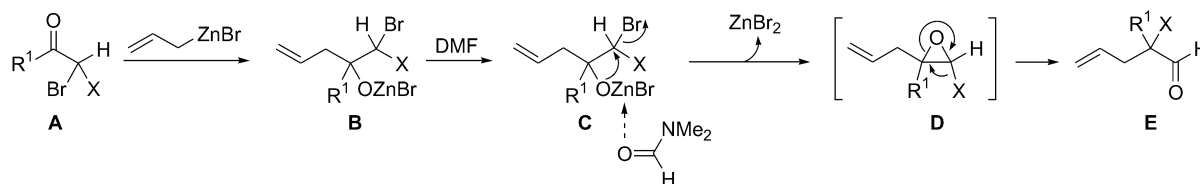
To the best of our knowledge, in conjunction with previously reported examples,^[3,4,15] we propose a possible mechanism depicted as follows (Scheme 1). Firstly, α,α -dihalo ketone (**A**) leads to the intermediate **B** upon

Table 4 Reactions of α,α -dihalo-ketones with substituted allylic zinc reagents^a

$\text{R}-\text{C}(=\text{O})-\text{CBr}_2 \xrightarrow[\text{(2) DMF 4 h}]{\text{(1) R}^1\text{ZnBr, THF, 30 min}} \text{R}-\text{C}(\text{Br})(\text{R}^1)-\text{CHO}$				
Organozinc reagent	R	Product	<i>dr</i> ^b	Yield ^c /%
	C ₆ H ₅	 4a	>99 : 1	32
	<i>p</i> -ClC ₆ H ₄	 4b	>99 : 1	30
	<i>p</i> -BrC ₆ H ₄	 4c	>99 : 1	35
	C ₆ H ₅	 4d	>99 : 1	27
	<i>p</i> -BrC ₆ H ₄	 4e	>99 : 1	30
	C ₆ H ₅	 4f	≈3 : 5	23
	<i>p</i> -BrC ₆ H ₄	 4g	≈2 : 3	25

^a Reaction conditions: α,α -dihalo-ketones (0.5 mmol), allylzinc bromide (1.0 mmol), THF (3 mL) and DMF at room temperature under nitrogen for 4 h. ^b Diastereomeric ratio (*dr*) determined by ¹H NMR analysis. ^c Isolated yield is based on α,α -dihalo-ketones after silica gel chromatography.

Scheme 1



undergoing the allylation. Then DMF is added to the solution of **B** and the oxygen atom of ligand DMF will coordinate with the Zn atom of the intermediate to activate it. Finally, the intermediate **B** undergoes intramolecular nucleophilic substitution to provide an α -haloepoxide **D**, which is subsequently converted into the desired product (**E**).

In conclusion, we have described a convenient and efficient method for the preparation of a series of multifunctional carbon compounds namely, α -halo, α -allylic aldehydes, using four types of allylic zinc reagents. Secondly, the study on its generality and scope demonstrated here clearly suggests the usefulness and versatility of our method. Future optimization on the reaction conditions is strongly encouraged to obtain higher reaction yields.

Acknowledgement

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